

DLR-TomoSAR — Physics-constrained learning

five loss terms · curriculum · the unrolled solver

July 2026

Why constrain the solution space

- The labels are classical fits: the network's ceiling is the label quality — and Act IV showed the network happily leans on statistical shortcuts.
- Physics terms restrict predictions to profiles **consistent with the SAR signal model**:
 - today — as a regularizer: physical quantities recomputed from the predicted profile must match those of the GT profile;
 - next — as self-supervision against the *measured* covariance itself, removing the label ceiling entirely.
- Five terms implemented and theory-tested; every one differentiable end-to-end through the Gaussian-mixture curve reconstruction.

The shared forward operator

All terms use the vertical wavenumber and the steering matrix on the elevation grid:

$$k_z^{(i)} = \frac{4\pi b_i}{\lambda r_0 s}, \quad A_{in} = e^{j k_z^{(i)} \xi_n}$$

- Geometry resolved from campaign track parameters — not nominal defaults: per-pass perpendicular baselines, look angle, slant range.
- **Per-pixel k_z field** for range/azimuth-varying geometry, built at preprocessing and threaded dataset → trainer → loss.
- Steering sign settled empirically ($+k_z$: median peak error 0.67 m, within one Rayleigh cell).

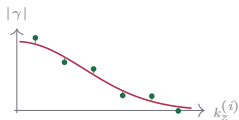
Data-consistency terms (the ranked top two)

Coherence re-synthesis

coherence domain

$$\gamma_P(k_z^{(i)}) = \frac{\sum_n P(\xi_n) e^{jk_z^{(i)} \xi_n}}{\sum_n P(\xi_n)}$$

$$\ell_{\text{coh}} = \left\langle \frac{1}{N_s} \sum_i |\gamma_P(k_z^{(i)}) - \gamma_T(k_z^{(i)})|^2 \right\rangle$$



γ_P from the predicted mixture pulled onto the target coherences, baseline by baseline.

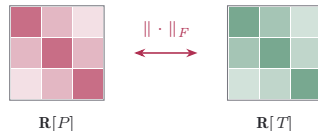
Cloude 2006 (PCT); same Fourier kernel as gamma-Net (Qian et al.).

Covariance matching

covariance domain

$$\mathbf{R}[P] = \mathbf{A} \text{diag}(P) \mathbf{A}^H \Delta \xi$$

$$\ell_{\text{cov}} = \left\langle \frac{\|\mathbf{R}[P] - \mathbf{R}[T]\|_F^2}{\|\mathbf{R}[T]\|_F^2} \right\rangle$$



Synthesised vs target covariance — structure and relative scale, entry by entry.

COMET (Ottersten, Stoica, Roy 1998); TomoSAR: Cros & Ferro-Famil 2025; CS fidelity: Berenger et al. 2023.

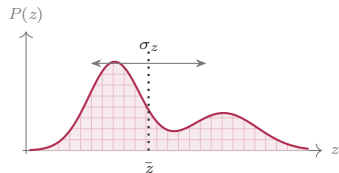
Descriptor terms: moments and total power

Moments

mass · phase-centre $\bar{z}$ · spread σ_z

$$m = \sum_n P(\xi_n) \Delta \xi$$

$$\bar{z} = \frac{\sum_n \xi_n P(\xi_n)}{\sum_n P(\xi_n)}, \quad \sigma_z^2 = \frac{\sum_n (\xi_n - \bar{z})^2 P(\xi_n)}{\sum_n P(\xi_n)}$$



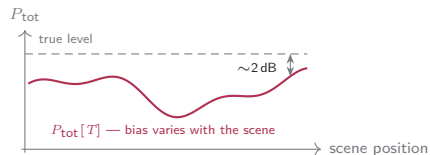
Centroid and spread are bias-robust *ratios* (Cros & Ferro-Famil 2025).

Total power

the one absolute-power quantity

$$P_{\text{tot}}[\cdot] = \sum_n (\cdot)(\xi_n) \Delta \xi$$

$$\ell_{\text{pow}} = \left\langle (P_{\text{tot}}[P] - P_{\text{tot}}[T])^2 \right\rangle$$

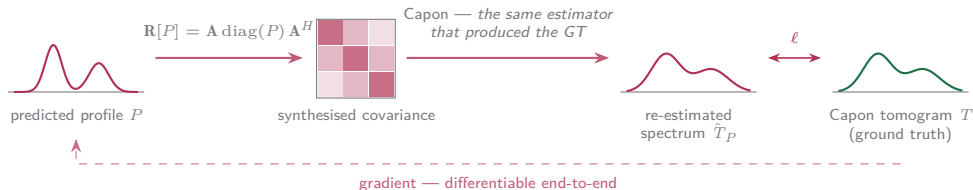


The target $P_{\text{tot}}[T]$ is not the true power: its bias moves with scene content (Lopez-Dekker & Mallorqui 2010) — the loss would imprint that bias.

The estimator-cycle term: Capon cycle

Push the prediction through the *same estimator* that produced the ground truth:

$$\hat{T}_P(\xi_n) = (\mathbf{a}^H(\xi_n)(\mathbf{R}[P] + \epsilon\bar{\sigma}\mathbf{I})^{-1}\mathbf{a}(\xi_n))^{-1}$$

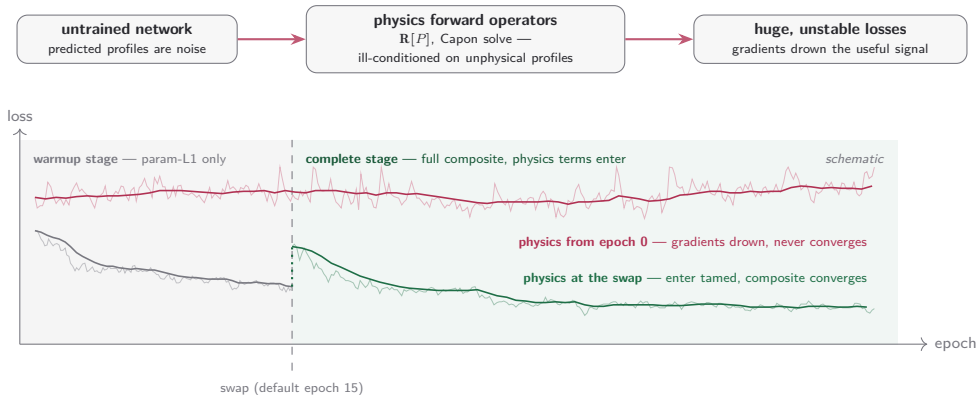


- Absorbs the Capon bias and the PSF **by construction**.
- Costs an $N_s \times N_s$ solve per pixel — the most expensive of the five.

Cycle-consistency lineage: Yaman 2020, Lu 2021, Vella & Mota 2020, Angermann 2023.

Why the physics terms need a curriculum

The physics terms are the most sensitive consumers of the reconstructed profile — and at random initialisation they see only noise:



Swap mechanics, calibration — and status

Each detail is an anticipated failure mode:

- **Resets at the swap:** LR schedule origin, LR warmup, Adam moments, early-stopping counters — optimizer state calibrated to the old loss is stale under the new one.
- **Checkpoint-baseline reset:** best-loss tracking restarts across the non-comparable loss scales — otherwise the pre-swap best could never be beaten.
- **Stage inheritance:** the complete stage is the base configuration, warmup a derived pre-phase materialised at launch — the stages cannot drift.
- **Loss-scale probe:** every active term run at unit weight over probe batches, IQR-filtered, folded into the per-term weights — physics terms enter at calibrated magnitude, not guessed weights.

Status

Implemented, SAR-theory-verified, geometry wired end-to-end — the single-term $K=2$ sweep is in (next frames); the Capon-cycle arm, the most expensive of the five, is still queued.

The single-term sweep — every family helps low, two collapse high

- At $w \leq 0.05$ **every family beats the no-physics baseline** (0.620) — adding a physics term at gentle weight essentially never hurts.
- **Coherence re-synthesis peaks at $w=0.25$** (0.944, the sweep's best arm) — the theory-ranked #1 term is also the empirical #1.
- **Moments is the robust family:** above baseline at every weight, no collapse anywhere on the grid.
- **Covariance matching and total power collapse past $w=0.25$** — the two absolute-scale terms overpower the parameter loss (curve R^2 drops to 0.72 at $w=1$).
- The Capon-cycle arm is not in this sweep — queued, costliest term.

term \ w	0.01	0.05	0.25	0.5	1
coherence resyn.	0.819	0.779	0.944	0.609	0.739
moments	0.855	0.876	0.792	0.861	0.767
covariance match	0.834	0.871	0.578	0.306	0.187
total power	0.789	0.817	0.689	0.491	0.144
no-physics baseline	0.620				

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=2$ · presence A · full norm ladder · hv flips · patch 64×32 · full stack $\rightarrow$ both trunks; loss = param-L1 + 0.5 cosine curve + one physics term at weight w ; five seeded replicas per arm, held-out test. Cell = grouped score: equal-weight mean of the five metric-group scores (curve error / variance explained / structural similarity / shape / peak location), min-max scaled *within this 21-arm comparison* — a rank statistic, no seed spread. **bold** = best in row, marked only where best and worst differ by more than 5% (every row clears it); below baseline ; sweep best . Val losses are not comparable across arms (each optimises a different composite).

The verdict — physics terms buy tail insurance, not headline gains

- **Headline gains are small:** curve R^2 $0.747 \rightarrow 0.755$ at the best arm — at the edge of combined seed noise; SSIM moves in the fourth decimal on all three axes.

- **The real effect is the per-pixel tail:** the baseline's mean pixel R^2 is -1.17 , wrecked by catastrophic pixels ($p_1 = -14.9$). The descriptor and covariance terms cap the blow-ups and pull the mean positive.

- **Physics arms also stabilise seeds:** covariance $w=0.05$ runs at a fifth of the baseline's curve-metric seed spread.

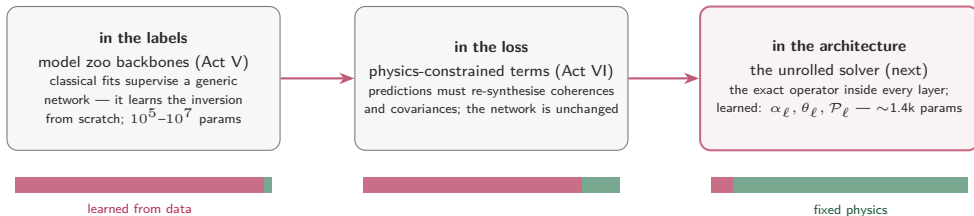
- **Detection and localisation sit within seed noise** everywhere — the terms shape the profile, not the scatterer set.

metric	baseline	coh. 0.25	mom. 0.05	cov. 0.05	pow. 0.05
curve R^2 $\uparrow$	0.747 $\pm$.006	0.755 $\pm$.007	0.755 $\pm$.002	0.748 $\pm$.001	0.751 $\pm$.005
PSNR [dB] $\uparrow$	52.63 $\pm$.10	52.78 $\pm$.13	52.76 $\pm$.04	52.65 $\pm$.02	52.70 $\pm$.08
pixel R^2 mean $\uparrow$	-1.17 $\pm$.79	-0.88 $\pm$.53	+0.53 $\pm$.19	+0.77 $\pm$.01	+0.73 $\pm$.05
pixel R^2 p_1 $\uparrow$	-14.9 $\pm$ 11.4	-12.4 $\pm$ 4.0	-2.2 $\pm$ 1.1	-0.5 $\pm$ 0.1	-0.9 $\pm$ 0.4
pixel R^2 min $\uparrow$	-13.2k $\pm$ 3.3k	-12.2k $\pm$ 4.8k	-4.4k $\pm$ 1.2k	-4.2k $\pm$ 0.2k	-4.1k $\pm$ 0.1k
cos med $\uparrow$	0.965 $\pm$.004	0.966 $\pm$.002	0.966 $\pm$.002	0.966 $\pm$.002	0.966 $\pm$.001
recall $\uparrow$	0.901 $\pm$.001	0.903 $\pm$.002	0.902 $\pm$.001	0.902 $\pm$.001	0.900 $\pm$.001
precision $\uparrow$	0.895 $\pm$.008	0.892 $\pm$.009	0.896 $\pm$.005	0.899 $\pm$.005	0.898 $\pm$.003
count exact $\uparrow$	0.908 $\pm$.007	0.909 $\pm$.005	0.910 $\pm$.004	0.909 $\pm$.004	0.908 $\pm$.002
μ MAE $\downarrow$	1.71 $\pm$.07	1.72 $\pm$.06	1.69 $\pm$.04	1.65 $\pm$.02	1.67 $\pm$.03
σ MAE $\downarrow$	0.774 $\pm$.011	0.778 $\pm$.011	0.771 $\pm$.018	0.754 $\pm$.007	0.761 $\pm$.005
a MAE $\downarrow$	0.182 $\pm$.004	0.182 $\pm$.005	0.182 $\pm$.003	0.187 $\pm$.005	0.184 $\pm$.004
peak err mean $\downarrow$	1.231 $\pm$.016	1.216 $\pm$.008	1.221 $\pm$.010	1.213 $\pm$.007	1.221 $\pm$.010

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=2$ · presence A · full norm ladder · hv flips · patch 64×32 ; loss = param-L1 + 0.5 cosine curve + term at w (column = each family's best-scoring weight); five seeded replicas per arm, cells mean $\pm$ seed std; held-out test, 5.0 M px. Curve MAE and three-axis SSIM are flat across the five arms (MAE 0.0365–0.0370; SSIM elev/range/azim 0.9671–0.9675 / 0.9510–0.9521 / 0.9610–0.9616). **bold** = best in row, marked only where best and worst differ by more than 5% — only the pixel- R^2 tail rows clear it; every other row sits inside combined seed noise. Shading keys each block to its like-coloured bullet. recall / precision / count matched (Hungarian pairing); pixel R^2 min in thousands (k); μ , σ and peak err in elevation units, a in linear amplitude; cos med: median per-pixel cosine to the GT profile.

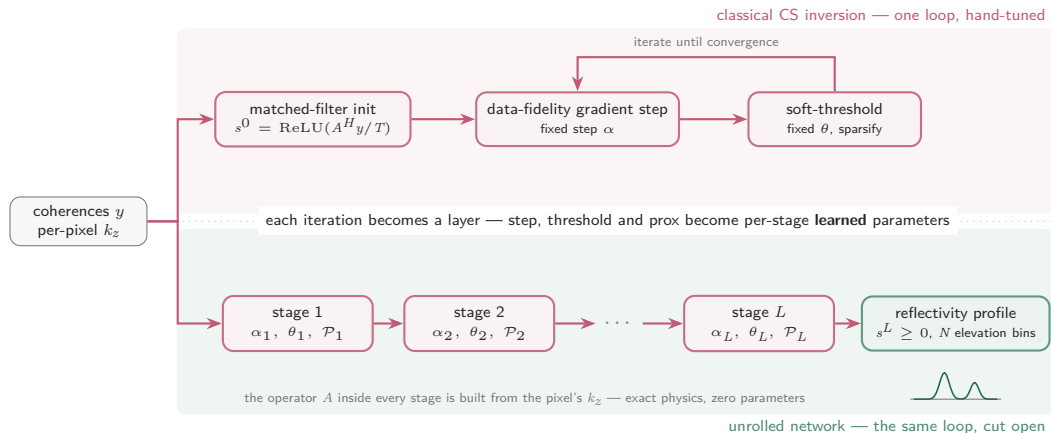
Context: where does the physics live?

Three places the SAR signal model can enter a learned inversion — the project now covers all three:



The unrolled family probes the far end of the axis — **the question it answers:** how much of the zoo's capacity is spent re-learning physics we already know?

The unrolled model: the solver becomes the network



$$r^\ell = s^\ell + \alpha_\ell \text{Re}[A^H(y - As^\ell)] / L_{\text{lip}}, \quad s^{\ell+1} = \text{ReLU}(\mathcal{P}_\ell(r^\ell) - \theta_\ell)$$

γ -Net lineage (Qian et al., IEEE TGRS 2022); LISTA (Gregor & LeCun 2010); ISTA-Net (Zhang & Ghanem 2018). $\mathcal{P}_\ell$: per-pixel residual 1D convolution along elevation; $L_{\text{lip}} = TN \text{d}z^2$ normalises the step. Default $L=8$: $\sim 1.4\text{k}$ parameters.

Adapting γ -Net — a different unknown, a per-pixel operator

γ -Net — one scene geometry, weights learned



..... a fixed matrix can be learned — an operator that changes at every pixel cannot



this project — per-pixel geometry, physics exact, scalars learned

γ -Net inverts $g = \mathbf{R}\gamma + \varepsilon, \quad \gamma \in \mathbb{C}^N$

this project inverts $y = A(x)s, \quad s \in \mathbb{R}_{\geq 0}^N$

g : one complex look per track (first-order); y : per-track coherences (second-order) — averaging interferograms cancels the speckle phase, so the recoverable object is the power profile s , not the complex reflectivity γ . ε : measurement noise; T : tracks; N : elevation bins; $\mathcal{P}_\ell$: per-pixel residual 1D convolution along elevation.

Adapting γ -Net — mechanism by mechanism

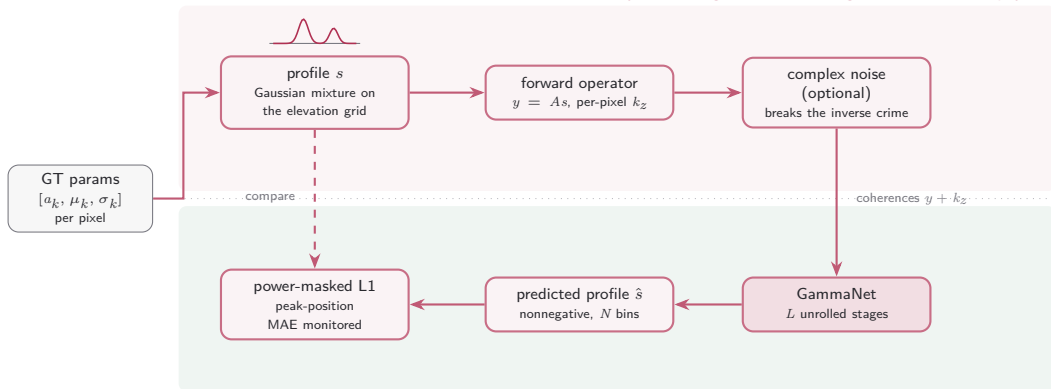
mechanism	γ -Net (Qian et al. 2022)	this project	why
iteration	gradient step $\rightarrow$ shrinkage, L stages	= kept	the unrolling premise itself
supervision	train on simulated measurements	= kept	the exact operator generates the pairs
unknown	complex reflectivity $\gamma \in \mathbb{C}^N$	$\rightarrow$ power profile $s \in \mathbb{R}_{\geq 0}^N$	speckle phase averages out of coherences
measurements	single-look stack g	$\rightarrow$ per-track coherences y	the pipeline is second-order end to end
operator	one fixed $\mathbf{R}$, scene-wide	$\rightarrow$ $A(x)$ from the pixel's k_z	airborne geometry varies across the scene
learned weights	$\mathbf{W}^\ell$ per stage, coupled to $\mathbf{R}$	$\rightarrow$ none — scalars α_ℓ, θ_ℓ	a per-pixel matrix cannot be learned
shrinkage	support selection + piecewise-linear function	$\rightarrow$ prox $\mathcal{P}_\ell$, then $\text{ReLU}(\cdot - \theta_\ell)$	the prior must respect $s \geq 0$
model order	Bayesian information criterion after inversion	$\rightarrow$ downstream parameter extraction	the pipeline already fits K scatterers

The contract every change preserves: **the physics stays exact inside the network — only the prior is learned.**

support selection: the largest-magnitude entries bypass the threshold at each stage; piecewise-linear function: γ -Net's learned replacement for soft-thresholding; coupled weights: $\mathbf{W}_2^\ell = \mathbf{I} - \mathbf{W}_1^\ell \mathbf{R}$ halves the free parameters per stage. γ -Net reaches its accuracy at 12 stages; default here $L = 8$. **green** = exact physics carried by the operator · **red** = learned · grey = kept from γ -Net.

Training the unrolled model against the exact operator

synthesis — ground truth through the exact forward physics



inversion — the unrolled network runs it backwards

Phase 1 (implemented): learn to invert the *exact* operator on synthesised coherences. Phase 2 (pending): measured covariances from the stack — no label ceiling.

Isolated family — own registry, trainer and entry point (`train_unrolled`); optimizer splits steps/thresholds from the prox; no AMP, the operator is complex-valued.

Why it exists: the physics-native comparison point — if $\sim 1.4k$ parameters of encoded physics match capacity-matched vision backbones, those backbones are mostly re-learning known physics.